THE WAR ON THE CRITICAL EDITION VOLUME 2

Piano Games: Multiple compositions from a single source

Kim Cunio

Queensland Conservatorium,
Griffith University

ABSTRACT

In my so called 'serious' research, (that into best practice realization of ancient and medieval music), a major theme has been the preparation of multiple realizations of a text or musical work, something that can be defined as music that has no critical or singular edition. This premise has had a profound effect on both my realized early music and new art music composition. This paper continues earlier research into the process of writing multiple compositions from a singular source. Previously I have written multiple compositions based on fragments of Hildegard of Bingen's *Ordo Virtutum* and sent a solo shakuhachi piece out to the public as a series of garageband loops, projects that stimulated discussion about the home space of composition. This part of the experiment is concerned with making new compositions from existing compositions without changing a note of 'the score'. In 2010 I wrote a series of simplistic piano preludes, recording them on a MIDI[1] instrument and treating those recordings as source data which could be exported as either a score or a MIDI file. Experimentation was then undertaken to see if new compositions can be realized wholly within a commercially available DAW.[2] Each original work has given rise to two variant compositions, which utilize some of the techniques of MIDI manipulation in screen, art and popular music styles.

1. INTRODUCTION

The process of recomposing is hardly new. It can even be argued that until the rise of the named composer that the majority of music was constantly recomposed, something that still occurs in many of the world's oral traditions. The processes of notation and transcription play a part of the solidification of music, allowing certain versions of a musical form to gain currency as the norm, whether in the preservation of historical editions or in performance practice itself. I described this in the first part of this project:

> The critical edition is at the heart of western art music. When someone asks to hear *Ave Maria* at their wedding there is an inherent cultural assumption that they will hear a particular version of the *Ave Maria*, usually composed by Schubert or Bach/Gounot. Though the Bach/Gounot is an arrangement of an earlier critical edition it has been absorbed into the cannon of western art music alongside the Schubert, and as such either composition can be attributed to a composer(s). Musicology can document the works of western composers thanks to the invention of notated music by Guido of Arrezzo in the 11th Century, and the extant works of the thousands of composers who have worked in both sacred and secular music. These thousand year old notations give us many of the best and worst aspects of our current musical life: they allow copyright and patents to flourish, intellectual capital to be recognised, but they also allow singular editions to push their multiple counterparts to the side. For example an artist may play a hit song many different ways but the critical edition is always the recorded and disseminated version of the work. (Cunio, 2009).

Recording has revolutionised the dissemination of music and also the way it is delivered. In recent times it has had a profound effect what actually constitutes a critical edition. A century later it seems impossible to imagine a world without music on demand. Though recording has signaled many changes in culture the most pertinent for this project is the singularization of the recorded work. A century ago a score was the form in which people held a critical edition, and though in itself a singular item the score was open to interpretation in reading and playing. Today for most young people music is a download, critical edition often devoid of its subtext and history. Instead of hearing many versions of a piece of music as innately equal, as with the live performances of established repertoire such as opera or jazz standards, one recording is often both the composition and dissemination of a piece of music.

It is common for there to be no formal score in a contemporary piece of music. Sometimes a score is even prepared retrospectively, particularly with popular and screen music, processes dominated by composition on an instrument or MIDI keyboard. The journal *Recent Researches in the Oral Traditions of Music* defines this point of change:

[1] Musical Instrument Digital Interface, a standard for transferring digital music related data.

[2] Digital Audio Workstation, a computer based recording, editing and sequencing program.

> Recent Researches in the Oral Traditions of Music encourages scholars to rethink the critical edition as a crucial component in the current rapprochement between ethnomusicology, historical musicology, and cultural studies. As new media make it possible to experience musics from throughout the world, as oral traditions have become essential to the globalization of local musical practices, and as popular musics give postmodern meaning to historical diasporas, so too does Recent Researches in the Oral Traditions of Music invite music scholars to conceive of editions that will contribute fundamentally to some of the most critical debates of our day. (Bohlman, 2005).

While the computer has changed the game for music making and recording, art music composers and institutions are quickly coming to terms with the ramifications. The digital transfer of music offers a potentially perfect copy of a performance that can be transcribed or learnt orally, making it a meeting place between oral and written forms. It can be argued that notation, as we historically understand it, is now one of a number of processes to preserve and record music. MIDI, the DAW and wave file composition (whereby the composition takes place after the recording of the individual parts), have replaced traditional scoring for many composers. In addition to this we now extend the term composition and composer well beyond the historical western definitions. The composer of a work does not necessarily have to know the craft of notation, nor be able to perform a work the same way twice. Reid (2007), states that a written score can range from a chord chart to a Pro Tools file, yet a standard composition degree at a tertiary institution is still mainly concerned with the authorship of singular, critical written works. Ofcourse such things are commonplace in moving image work. The composer Hans Zimmer not only developed his own sample library to replace the traditional orchestra, he has invented a hybrid sound of natural and synthetic sounds that are as much a part of the score as the notes, as these fusions sounds epitomize the thickness of the modern Hollywood sound. (Davies, 2011).

This paper therefore serves two functions: to document my developing practice based research as well as stimulating discussion into the processes and ramifications of recomposing around existing material, something that also applies to my early and intercultural music (Cunio, 2009b). Earlier in this research I recomposed around the Prologue from Hildegard of Bingen's *Ordo Virtutum*. Although new works were generated from a critical edition, it was clear that the original work still lay with the historical author and that these derivative works could be seen as some people as merely constituting an arrangement or realization. In this part of the process the notion of a home music is completely subverted, multiple compositions take place from a singular source, with the essential score data unchanged.

2. PIANO GAMES

2.1. Preamble

The genesis of *Piano Games* lies itself in technology. I was commissioned to work with the Australian author Dr Stephanie Dowrick on a CD of meditation commentaries in 2009.[3] At the initial meeting it was decided that two new pieces of music would be written in addition to an underscore for the spoken word. At this point I mentioned that I had recently updated my piano sample library and proceeded to play a short prelude that I had rendered on a virtual Fazioli piano. At this moment I had a small epiphany, realizing that for most people music technology still constituted pedals, boards and electronics, that the recent advances in synthesis and sampling have gone almost unnoticed despite their ubiquitous presence in television, film and popular music.

Although I own a 75 year old Mignon grand piano and have access to all the resources of the Conservatorium I decided at that moment to 'smuggle in' a sampled piano and cello to this commission, and pass the samples off as real players to see if anyone noticed. This was the birth of *Open Mind*, the first piece in this project. The piano part was recorded on a $200 plastic keyboard after which editing work was undertaken in a MIDI editor. The MIDI data from the performance was edited in relation to attack, decay, velocity and pedal duration, summed in the analogue domain and processed through the Bricasti M7, a newly released hardware digital reverb. After this the audio was parallel compressed to suggest the feeling of attack present in a real player. A similar process was undertaken with the cello part. A simple line was composed and played on the MIDI keyboard and different cello patches were alternated, primarily between up and down bows (the latest sample libraries now do this automatically). Extensive equalization was utilized to open the sound of the lowest notes of the cello, and provide greater bass response to the higher notes (a longstanding problem in many sample libraries). After this the part was summed in the analogue domain, limited, and again processed with the Bricasti reverb. I invented a pseudonym for the cello player,[4] and delivered the work to the company. Not a word was spoken, everyone from the company loved the work, and although it was obviously a 'fake' to me, I delighted in the realization that no-one even noticed. I then wished to extend the experiment to see if I could make new works from the one score.

[3] *Meditation Essentials,* Stephanie Dowrick and Kim Cunio: Vision Australia 2009 ISBN: 9781864826210.

[4] Eliyahu Elias, two iterations of my middle name.

2.2. Preludes

In order to proceed with the project I needed to write some source material. In keeping with the notions of subversion that constituted the birth of the project a series of structured improvisations were entered into to create new source material. Each piece would be both a prelude in the historical sense of the word, and a prelude to re-composition. The first two pieces presented themselves with ease. I exported the piano part of *Open Mind* as the first prelude and while selecting it found a sound check in the previous take, which I also selected. This piece was quickly named *Sound Check*, and the project had started. From these two works a series of parameters were developed for additional source composition:

- That each prelude be written in a particular scale.
- That modulation be kept to a minimum.
- That each prelude be capable of being notated, played acoustically or re-improvised at a later point.
- That each prelude be played into a DAW allowing the computer to store MIDI attack, decay and velocity data.
- That this project be undertaken with one commercially available DAW, in this case Digital Performer, with the Fazioli piano sample on East West's Goliath virtual instrument.5

In total six source preludes were created for which summary data is below. These works constitute CD1, with the derivative works constituting CD2 (Figure 4).

Name	Length	Key [6]	Style
Sound Check	2:55	a	Slow Persian Hijaz scale
Baby Bach improvisation	6:48	g	Contrapunctal
First Species	4:29	c	First species counterpoint
Lydia improvisation	6:16	D	Lydian mode
Ambi	6.43	e	Improvised ostinato
Open Mind	5:12	g	Filmic improvisation
Total	32:21		Improvised piano music

Figure 1. Summary data for Preludes used in Piano Games.

[5] A software based sampler loaded as an instrument track into a DAW.

[6] Capital letter denotes major key.

In a traditional music project such improvisations would probably be scored either through direct transcription or through the exporting of the MIDI data to Finale. In this project they were not altered, leaving the MIDI data looking decidedly non idiomatic on the QuickScribe window in Digital Performer.[7] The reasons for this are telling, a direct transcription would privilege the source prelude beyond its intended significance, subverting the later works. Resisting the score in this case is also tantamount to resisting the critical edition. The example below is typical of the score data in the project.



Figure 2. Sample of QuickScribe notation in *Sound Check.*

2.3. Play

The ensuing process is one that I as a researcher rarely have the luxury of entering into. I described it as a process of 'rampant experimentation' in a lecture to Music Technology students at the Queensland Conservatorium in April 2011. (Cunio, 2011). I started this phase with the objective of seeing how many fundamental parts of a composition I could manipulate without changing the score, and within a short period of time the list was extensive, as can be seen in Figure C. I was looking to manipulate this data in a mainstream DAW as opposed to a processing program such as Max MSP or Pure Data. Even though the commercial DAW is not designed for such re-composition there are certain advantages in staying within such a platform. They include time saved on writing and debugging new systems, the inherent audio quality of the commercial DAW, the integration of 3rd party products in regards to latency compensation and the availability of plug ins and the ease in which virtual instruments can be added to the initial MIDI data. There is something far more important than the method in this project however, which is the inspiration of the act of composition. Within a short period of time this project took a life of its own; processing tools replaced traditional tools and I was writing freely.

Objective	**Method**
Change instruments	Assign MIDI track to new virtual instrument
Change tempo	Change global tempo on project

[7] This is a very common issue in music for moving image where budget and timelines often do not allow for an adequate score to be prepared, leaving some session musicians with the unenviable talks of decoding a poor DAW score in a recording session. (Lee, 2009).

	Insert tempo changes into conductor track
Change key	Transpose plug in
Harmonization	Transpose plug in, chains of transpositions
Reassign scales	Humanize plug in to create new pitch maps
Change performance	Traditional MIDI dialogue tools
	Humanization plug in to control onset,
	duration on velocity, off velocity
	Time shift plug in to shift bars and beat data
Change rhythm grooves	Quantize / groove quantize, new DNA
Arpeggiation	Arpeggiator to control subdivision,
	note order, range
Delay repeats	Echo plug in to control subdivision and
Invert pitches, key	Invert pitch plug in
	Nomination of central pitch and inversions
Remap Midi data	Reassign CD plug in

Figure 3. Manipulation of initial Prelude data in Piano Games.

In this from of manipulation several things quickly became cognizant. The order of manipulation would effect the final outcome of the MIDI data just as the order of inserting plug ins to audio effects its transmission in a DAW. A number of the MIDI plug ins also had randomize functions which move the project into the realm of algorithmic composition; and the order in which data manipulation took place effected my perception of composition choices. Within a short period of time the project turned full circle. I had constructed it to a very strict brief so as to manage its scope, but it still looked too vast. Further limitation was needed in order for the project to have a unified theme and outcome. To respond to this challenge a number of parameters were developed:

- To work wholly within a portable computing environment without the aid of extensive studio equipment, analogue processing or large rack units of virtual instruments. This included bouncing final mixes from within bus tracks.
- To limit myself to four commercially available software virtual instruments, Modartt's Pianotech, a synthesis piano for piano, electric piano vibraphone, marimba, and celeste, the Wallander synthesis instrument for brass and woodwind, the Garritan instruments for Finale for strings and additional percussion and the MOTU Polysynth for synthetic pads and drones. Each of these was chosen for ease of use, price point and low CPU usage.
- To try and limit myself to two new compositions from each source Prelude.
- To not formally compose or notate a re-composition. In other words to keep each new work as an ephemeral act, undertaken in a short period of time.
- To not change a note of the initial score data or copy a plug in chain from one set of source data to another.

Once familiar with this set of parameters the task of writing music was familiar. As with any act of creation I had to find a balance between planning and intuition, or in this case prescriptive and reactive editing. Prescriptive editing in this project is defined as a deliberate manipulation to achieve a predetermined compositional outcome, whereas reactive editing is defined as a process of responding to the source data by improvising parameter changes as the work is played, in the style of live electronic and coding performance. Needless to say it is in the combination of methodologies that art takes place. In all thirteen new compositions were written. Each prelude inspired two derivative compositions though one of the source pieces *Ambi,* gave rise to a third new work. A critical packaging decision was whether to combine the Preludes with the derivative works or to present them separately. It was decided to present them separately as the derivative works comprise a full length CD in themselves, and it is anticipated that the initial preludes will be either recorded acoustically or have a new improvised realization recorded at a future point. In this project the order of the new compositions is a direct inversion of the order of the preludes. Figure 4 presents summary data for the new compositions in *Piano Games*, with a brief descriptor inserted underneath each new work.

Prelude Time/key	**New work**	
Open Mind g	*The varying nature of wind*	6:14
	Process music for piano, percussion & winds	
Bb	*Moving out*	4:22
	Ostinato led jazz groove for piano duo	
Ambi D	*Night light*	4:38
	Lullaby for solo piano	
E	*Comprovising*	6:12
	Groove variation for solo piano	
Ambi b	*Part Pärt*	5:52
	Ambient work for piano & string orchestra	
Lydia	*Case Study 1976*	3:56
	Modernist work for piano, harp, wind & brass	

D	*Tinkerbell*	4:23
	Dance for two pianos, keyboard & celeste	
First Species	*Mind games*	2:37
	Aleatoric miniature for solo piano	
	Invention for virtual prepared piano	5:09
	Aleatoric prepared piano solo	
Baby Bach g	*Driving the marimba*	3:28
	Piano, keyboard and marimba trio	
	Flight of the whole tones	3:20
	Fantasy for piano, vibes, harp & winds	
Sound Check a	*Raqam Sab'a*	4:02
	Piano duet in Arabic Hijaz scale	
F#	*Kaddish*	2:59
	Soundscape for piano, gamelan and synthesis	

Figure 4. Summary data for derivative compositions in *Piano Games.*

2.4. Manipulations

13 new pieces of music were composed from the 6 source Preludes. Each of these had a process of improvisation, formal manipulation, reflection and realization. Some pieces are recognizable with the main manipulations being related to rhythmic emphasis or subdivision. Others however are completely unrecognizable, transformed in tempo, instrumentation, pitch mapping, tempo, arpeggiation, playing emphasis and harmonization. A theme of the project has been a desire to see if the original preludes can be conclusively mapped to their derivative works. Where the original tempo, pitch or rhythm have been preserved this is straightforward, but in other cases the boundaries are not so clear. For example *Night light*, (CD2 track 3), derived from *Ambi* (CD1 track 5), appears to originate from the prelude *Lydia* (CD1 track 4), as it is in the same key with only one note changed in the scale. It also has a similar emotional quality to *Lydia,* which was an intentional part of the composition process. *Driving the Marimba* (CD2 track 10) and *The varying nature of wind* (CD2 track 1), have a great deal in common despite originating from different source preludes. They are both in the key of g, have a subtle rhythmic insistence and feature dialogue between the piano percussion and winds. Therefore it is not necessarily clear, or should it be clear where the exact boundary lies between source and derivative music. Although the pieces in CD2 have been collated according to the original preludes to allow greater reference between source and derivation this is not necessary in an artistic sense.

In order to understand this process the prelude *Lydia* and its derived compositions have been selected for study. This is because the two new works created form this prelude are dramatically different in regards to style and aesthetics. *Lydia*, the original is a response to the early 20th Century; tonal with pedalistic textures and a modal scale, in this case the Lydian mode from which its name derives. At times *Lydia* suggests the sounds of free jazz, although its limited harmonic content also imbibes a feeling of process minimalism. Its first derivative is *Case Study 1976*, a modernist work for prepared piano, harp, euphonium and tuba. *Tinkerbell*, the second derivative is a vibrant, ostinato based dance piece for two pianos, electric piano and celeste. *Tinkerbell* is immediately recognizable as it shares the same key and scale as the original, but *Case Study 1976* moves beyond the realm of rearranging the existing notes and constructs a new scale from the source data, which is combined with arpeggiation and an extended range in the piano and harp.

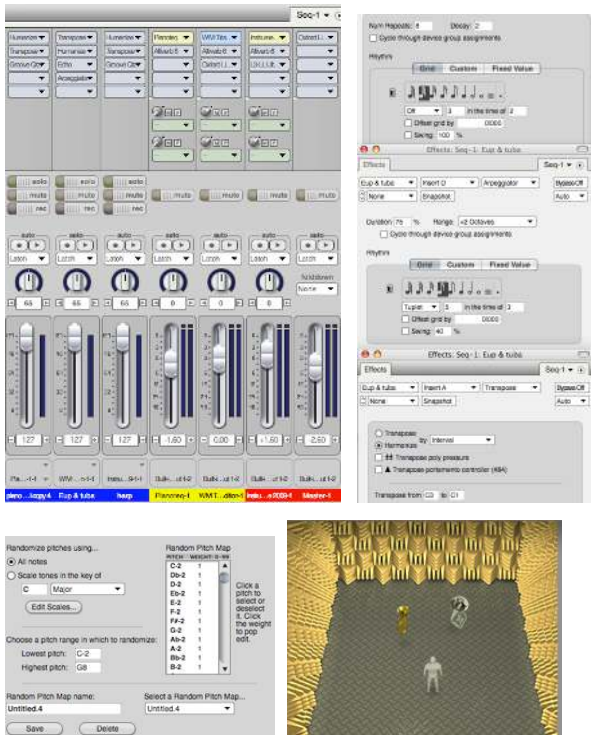


Figure 5. Example of mixing board, MIDI inserts and virtual instruments used in the production of *Case Study 1976.*

In this composition the prelude *Ambi* has been copied twice and assigned to different instruments, which in turn trigger virtual instruments. The score tracks trigger an iteration of Pianoteq for solo piano, the Garritan Personal Orchestra for harp, and the Wallander for euphonium and tuba. In this piece 10 MIDI inserts have been substantiated. Figure E also illustrates the MIDI processing used in the Wallander, which triggers both the euphonium and tuba. Seen from the top of the middle column in Figure E, they include a 32nd MIDI delay and an arpeggiator, which adds a range of two octaves to the

source sounded via a rhythmic filter set to a 5 over 3 tuplet. A harmonization of the initial data to two octaves below the source enables the Wallander to play different note combinations according to the ranges of the selected virtual instruments.

In the third column a chromatic pitch map allows all notes triggered by the Wallander, and therefore the arpeggiator and delay to be played in a randomly occurring chromatic scale. This is followed by a spatialization of the tuba and euphonium within a virtual anechoic chamber modeled in the Wallander. All the virtual instruments are then processed as audio tracks, a crude but effective way of modifying the sound of a virtual instrument. The Pianoteq is sent to a metallic drum reverb space as opposed to a concert hall, while the Wallander is sent to a small room and limited to allow greater presence and air in its 'performance'. The harp is sent to the same space as the piano with a greater tail and heavily limited to add some presence to the strings; something required to bolster the poor quality samples of the Garritan library. The tracks are then sent to a master track, which is then also limited. All the processing for this project has been undertaken on an entry-level laptop with 'in the box' plug ins available on any commercial DAW. Finally the score excerpt, shown in Figure 7 proves that the score has not been altered from the original prelude.

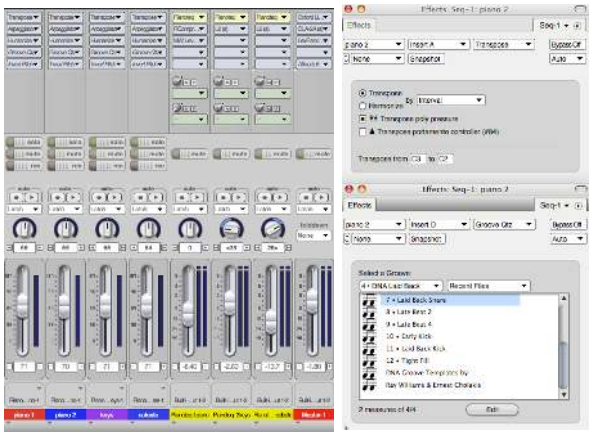


Figure 6. Example of mixing board and MIDI inserts used in the production of *Tinkerbell.*

In this composition the Prelude *Ambi* has been copied three times and assigned to different instruments, which comprise 3 different iterations of the Pianoteq plug in, one for piano, one for electric piano and one for celeste. 20 MIDI inserts are displayed in the mixer, of 14 have been substantiated, while each virtual instrument is sent to a limiter to avoid digital clipping. All the virtual instruments are also sent to a master track, which has the drafts of a limiter, compressor and two reverbs available for audition. The displayed MIDI plug ins display the Piano 2 track, transposing the data down a fourth and changing the rhythmic emphasis of the original to a programmed snare drum pattern. The score snapshot in figure 7 is taken from the Piano 1 and Piano 2 parts, which again shows that there has been no manipulation to the source data, as with *Case Study 1976.*



Figure 7. Identical score data from *Lydia*, *Case Study 1976* and *Tinkerbell*.

3. EVALUATION

Has the War been won? This is the question that beckons. There is little doubt that the process of re-composition in *Piano Games* is more radical than my previous projects, *Ordo Virtutum* by Hildegard of Bingen, and even *Namu Amida Butsu*, a solo work for shakuhachi that I exported as Garage Band loops. However, it must be asked if the resultant music is of sufficient quality to be classed as 'new composition'. A follow on question is whether this is a method of composition I might wish to use in the future, given its reliance on technology and its rigidly in regards to the initial score data. These are all subjects for ongoing debate that cannot be answered until the completion of this project. However to garner feedback I have addressed some of these issues in two lectures at the Queensland Conservatorium. In May 2011 I presented to the Music Technology Forum, a multi year undergraduate gathering of students, sessional lecturers and permanent staff. The response was mixed. While the music was broadly seen as having 'quality' the point was made that there might be a point in time in which the palette of manipulations was exhausted, where everything therefore sounded the same. Another respondent said that in order for this project to be conclusive the process should be undertaken on a number of audio platforms. (Cunio, 2011). Since that lecture a great deal of new music has been written in a number of music styles, which suggest that there is room for a significant amount of composition before the possibilities inherent in a DAW are exhausted, something I wish to explore in the future.

The research was also presented in August 2011 at the Queensland Conservatorium Higher Research Degree Colloquium, a gathering of Masters and Doctoral students that takes place once a week during the two semesters of the academic year. (Cunio, 2011b). In this presentation the response was again divided. A number of participants saw this as an arranging exercise, something akin to the tasks contained in composition technique tutorials. Other students applauded the work, seeing it as fresh, innovative and part of the melding of

popular and art music styles that is emerging in the 21st Century. It will be hard to gauge critical response until publication of this paper and dissemination of the accompanying CD, which should be completed in mid 2013. For this to happen there are still a number of decisions that must be made.

- Should the initial series of Preludes be realized on a Player Piano, played / improvised by a pianist and recorded, or presented 'as is'?
- Should the new compositions for piano be realized on a player piano, be mixed in a professional studio or presented 'as is'?
- In the case of realizations that have a significant algorithmic component should multiple realizations of the work be made before selecting a final 'new' composition?
- Should the MIDI files be made available to other composers to broaden the project?
- Is the musical material good enough to be released in a commercial environment? To be candid; does the music have artistic merit without the accompanying research?

In the meantime the results of this music will be placed in the web space of the QLD Conservatorium to enable comment, peer review and candid response, a major goal of the project.

4. REFERENCES

[1] Bohlman, P. (2005). General Editor. "*Recent* Researches in the Oral Traditions of Music" *A-R Edition 2005ISSN 1066-8209* introduction,.http://www.areditions.com/rr/rrotm.html.

[2] Cunio, K. (2006). "The Sacred Fire Scores", *Lotus Foot Music*, 2007, pp.3-40.

[3] Cunio, K. (2009). "The War on the Critical Edition Volume 1," in Proceedings of CreateWorld 2009: Mobile Me, Creativity on the Go, 30 November – 2 December 2009, Brisbane, Australia, 2009 pp. 14-20.

[4] Cunio, K. (2009b). "The Thread of Life: Intercultural music making and the process of defining cultural connection," in Proceedings of *MSA 2009*, 25-28 October 2009, Newcastle, Australia, 2009.

[5] Cunio, K. (2011). "Music Technology Forum – Lecture", QCGU, May 10, 2011, Brisbane, Australia.

[6] Cunio, K. (2011b). "Higher Research Degree Colloquium Presentation", QCGU, August 9, 2011, Brisbane, Australia.

[7] Davies, T. (2011). "Area of Strategic Interest Lecture in the Moving Image", QCGU, August 11 2011, Brisbane, Australia.

[8] Dowrick, S. & Cunio, K. (2009). "Meditation Essentials", Vision Australia, Sydney Australia, 9781864826210.

[9] Lee, H. & Cunio, K. (2007). "Hildegard Von Bingen, The Sacred Fire", *ABC Classics, Sydney Australia,* 4765705, 2007.

[10] Lee, H. (2009). "Recording for film and television", Personal communication 18 July 2009.

[11] Reid B. (2007). "Composition", Personal communication, 23 November – 15 December 2007.